Unemployment

Labour Economics, ECON207 — UG-II, 2026: Semester IV

Ravikiran Naik

April 2026

"It's a recession when your neighbor loses his job; it's a depression when you lose your own."

— Harry S. Truman

Sections Covered

§	Topic
12-2	Types of Unemployment
12-3	The Steady-State Rate of Unemployment
12-4	Job Search
12-5	Policy: Unemployment Compensation
12-8	Efficiency Wages
12-9	The Phillips Curve

Types of Unemployment

- **Frictional unemployment** arises when workers and firms need time to locate each other and to digest information about the potential job match.
- Even a **well-functioning competitive economy** experiences frictional unemployment, because some workers will unavoidably be "between" jobs.
- **Structural unemployment** arises when there is a mismatch between worker skills and the skills needed by firms, or a long-run imbalance between supply and demand.
- Structural unemployment is the **most concerning** type — the skills embedded in labour are no longer being put to productive use.

The Steady-State Rate of Unemployment

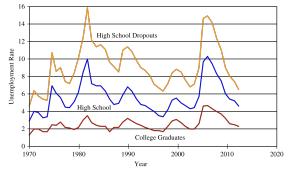
- The steady-state rate of unemployment depends on the **transition probabilities** among employment, unemployment, and the nonmarket sector.
- Let ℓ = the fraction of employed workers who **lose** their jobs and become unemployed.

- Let h = the fraction of unemployed workers who **find work** and get hired.
- In the steady state $\ell E = hU$, and since the Labor Force is $LF = E + U$, then
- The **Unemployment Rate**, $UR = \dfrac{U}{LF} = \dfrac{\ell}{\ell + h}$

Flows Between Employment and Unemployment

Suppose a person is either working or unemployed. At any point in time, some workers lose their jobs and unemployed workers find jobs. If the probability of losing a job equals l, there are l × E job losers. If the probability of finding a job equals h, there are h × U job finders.

FIGURE 12-3 Unemployment Rates by Education, 1970 to 2020

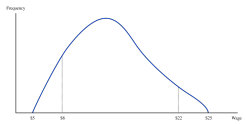


Source: BLS, Current Population Survey

Job Search

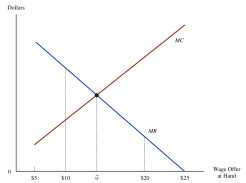
- The **asking wage** (reservation wage) makes the worker indifferent between continuing search and accepting the job offer at hand.
- An increase in the **benefits from search** raises the asking wage and **lengthens** the duration of the unemployment spell.
- An increase in **search costs** reduces the asking wage and **shortens** the duration of the unemployment spell.

The Wage Offer Distribution



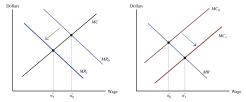
The wage offer distribution gives the frequency distribution of potential job offers. A given worker can get a job paying anywhere from $5 to $25 per hour.

Determination of the Asking Wage



The marginal revenue curve gives the gain from an additional search. The marginal cost curve gives the cost of an additional search. The asking wage equates marginal revenue and marginal cost of search.

Determinants of the Asking Wage

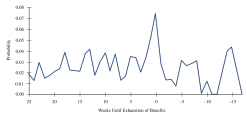


(a) A "present-oriented" worker has a high discount rate: MR shifts down, asking wage falls. (b) Unemployment insurance reduces search costs: MC shifts down, asking wage rises.

Unemployment Compensation

- Unemployment benefits are typically paid up to **26 weeks** of unemployment, but this length is frequently extended by Congress during recessions.
- The level of unemployment benefits depends on previous earnings. The **replacement ratio** averages about **60%** for low-wage workers and about **25%** for high-wage workers.
- Unemployment insurance **lengthens** the duration of unemployment spells and **increases** the probability that workers are laid off temporarily.

UI and the Probability of Finding a Job



Source: Katz and Meyer, "Unemployment Insurance, Recall Expectations, and Unemployment Outcomes," QJE 105 (1990): 973–1002, Figure IV

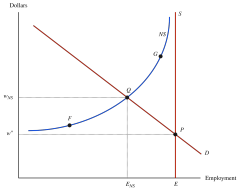
Why Would Firms Pay More Than They Have To?

- Some jobs are **hard to monitor** — firms cannot easily observe whether workers are putting in effort or shirking.
- If caught shirking, a worker gets **fired**. But firing is only a real threat if the worker's current wage is **better than the outside option**.
- Solution: pay **above the market-clearing wage** so that losing the job is genuinely costly.
- But if **every firm** does this, there are more workers willing to work than jobs available — unemployment is the result.

Efficiency Wages and Unemployment

- **Efficiency wages** arise when it is difficult to **monitor worker output**.
- Firms pay **above the market-clearing wage** to deter shirking: if caught, the worker loses a high-paying job, which is costly only if the wage exceeds the outside option.
- The above-market efficiency wage generates **involuntary unemployment** — workers willing to work at the going wage cannot find jobs.

The Determination of the Efficiency Wage



If shirking is not a problem, the market clears at wage w^*. If monitoring is expensive, the threat of unemployment keeps workers in line. The efficiency wage is given by the intersection of the no-shirking supply curve (NS) and the demand curve.

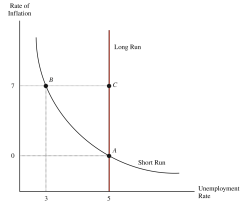
The Phillips Curve

- In 1958, A. W. H. Phillips found a **negative correlation** between inflation and unemployment in the UK (1861–1957).
- The idea: when unemployment is **low**, workers are scarce, firms bid up wages, and prices rise — inflation goes **up**.
- When unemployment is **high**, workers are abundant, wage pressure is weak, and inflation stays **low**.
- This suggests a **policy menu**: governments can "buy" lower unemployment by tolerating higher inflation, or reduce inflation by accepting more unemployment.

But Does the Trade-Off Last?

- The **short-run** Phillips curve slopes downward: surprise inflation temporarily reduces unemployment.
- **Why?** Workers do not immediately notice that prices have risen. Their reservation wage stays the same in nominal terms, but is now **too low** in real terms — so they accept offers faster.
- Once workers **catch on** and adjust their expectations, the reservation wage rises back and unemployment returns to the **natural rate**.
- In the **long run**, there is **no trade-off** — the Phillips curve is vertical.

The Short-Run and Long-Run Phillips Curves



How to Give Useful Feedback

- This is **your** chance to help me teach better — be **honest and fair**
- Think back over the **whole semester**, not just the last week
- Your grade is **yours** — don't judge the course by your performance
- Feedback should be **TO** me, not **ON** me — describe **specific behaviour**, not vague judgements
- Written comments matter most — say **what to keep** and **what to change**
- Not "this professor was awesome" — but *"the in-class activities helped me understand, not just memorise"*
- Reflect on **positives** and suggest **solutions**, not just complaints